CS 5330 Final Project Research

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CS 5330: Computer Vision & Pattern Recognition
Final Project Proposal
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Everything we considered for the project — datasets, simulators, challenges, and references — with a verdict for each. Deep dives live in datasets-and-simulators.md (logistics + gotchas), papers.md (full venue-verified paper list), and project-ideas.md (the seven project variants). Facts verified against primary sources 2026-07-21.

Datasets

Dataset	Sensor / domain	Size	3D labels	Role for us	Verdict
KITTI (object)	Stereo RGB + LiDAR, driving	12 GB	✓ 3D boxes	Outdoor eval (monocular-depth branch)	✓ **core**
SUN RGB-D	Real RGB-D, indoor	6.4 GB	✓ 3D boxes	Indoor eval (sensor-depth branch)	✓ **core**
Our own RealSense recordings	RGB-D, indoor	—	tape-measure GT	Metric accuracy + demo footage	✓ **extension #1**
nuScenes-mini	6 cameras + LiDAR, driving	3.9 GB	✓	Stretch: 360° map, center-distance metric, night/rain	~ stretch
nuScenes full	Full AV suite	314 GB	✓	Redundant with KITTI	✗ skip
nuImages	6 cameras, 2D only	large	✗	Detector fine-tuning only	~ training only
COCO	RGB, everyday scenes	—	✗	YOLO's pretraining source — not an eval set	(pretraining)
KITTI tracking	Driving video, track IDs	moderate	✓	Only for the dynamic-obstacle variant (idea 6)	~ variant
MOT17	Pedestrian video	moderate	✗ 2D IDs	Same — tracking variant only	~ variant
NYU Depth v2	Kinect, indoor	~4 GB	✗	Optional depth-model sanity check	optional
TUM RGB-D	Kinect + mocap poses	small	✗	Only for mapping/SLAM variants	✗ skip
ScanNet	RGB-D room scans	huge + signed TOS	✓ (mesh)	Only for the persistent-map variant (idea 3)	✗ skip
SemanticKITTI	LiDAR only	~80 GB	✗	No RGB-D alignment	✗ dropped

Dataset	Sensor / domain	Size	3D labels	Role for us	Verdict
Amazon AOT	Grayscale air-to-air	**13.4 TB**	✗ 2D	Airborne detect-and-avoid — wrong domain	✗ drop

- **KITTI** — the standard autonomous-driving benchmark suite (CVPR 2012), recorded from a car in Karlsruhe with stereo color cameras, a 64-beam LiDAR, and GPS/IMU.
  - Object subset: 7,481 training frames with 3D boxes for Car/Pedestrian/Cyclist; depth ground truth is sparse LiDAR projected into the camera — there is no depth camera.
  - Downloadable without registration from AWS Open Data (`s3://avg-kitti`) or the Kaggle mirror (per-folder download); CC BY-NC-SA 3.0.
  - The same "KITTI Vision" suite also hosts tracking, stereo, odometry, and depth-prediction benchmarks.
- **SUN RGB-D** — 10,335 indoor RGB-D frames captured with four real depth sensors (Kinect v1/v2, Xtion, RealSense) and annotated with 64,595 oriented 3D boxes (CVPR 2015).
  - The standard 3D-detection protocol is 10 furniture classes scored at mAP @ 0.25 3D IoU.
  - Official label tooling is MATLAB; the pre-extracted HF mirror provides the same files without it. Direct HTTP download, no registration.
- **Our own recordings** — not a public dataset: `.bag` captures from a RealSense D435-class camera (~1 GB/min) — tape-measured static scenes, handheld walkthroughs, and staged approach clips (pyrealsense2 wheels work on Ubuntu 24.04).
- **nuScenes** — 1,000 twenty-second driving scenes from Boston and Singapore (CVPR 2020): 6 surround cameras (1600×900), LiDAR, and radar; 1.4M 3D boxes over 23 classes; includes night and rain.
  - The mini split (10 scenes, 3.9 GB) is a strict subset with a direct download URL, no account; full trainval is ~314 GB (~60 GB keyframes-only).
  - Its official detection metric matches predictions by center distance (0.5–4 m thresholds), not box IoU.
- **nuImages** — 93k driving images with 2D boxes and instance segmentation masks; no depth, no LiDAR, no 3D annotations.
- **COCO** — 330k images of everyday scenes with 80 object classes in 2D boxes and masks; the dataset YOLO detectors are pretrained on.
- **Amazon AOT** — Prime Air's airborne detect-and-avoid dataset (2021 AIcrowd challenge): 4,943 flight sequences, 5.9M grayscale air-to-air images, 2D boxes only, 13.4 TB on AWS Open Data.
- **KITTI tracking / MOT17** — driving and pedestrian video benchmarks with ground-truth track identities; MOT17 ships public detections with its 7 train + 7 test pedestrian sequences.
- **ScanNet** — 1,500+ RGB-D room scans with per-frame camera poses and 3D instance labels on the reconstructed meshes; access requires a signed terms-of-use agreement.
- **TUM RGB-D** — Kinect sequences with motion-capture ground-truth camera trajectories, built for SLAM/odometry evaluation; no object labels.
- **NYU Depth v2** — 464 indoor Kinect scenes with 1,449 densely labeled frames (2D semantic segmentation); the classic indoor depth-estimation benchmark.

- **SemanticKITTI** — per-point semantic labels over KITTI's LiDAR sequences; LiDAR point clouds only, no camera-aligned depth.

Simulators

Simulator	Status (mid-2026)	Install on our setup	GPU load	Verdict
Habitat-Sim	Active	conda; ReplicaCAD instant, HM3D free request	Light	✓ **chosen** (extension #2)
Gazebo + ROS 2 + Nav2	Active; official Jazzy support	apt	Light	runner-up — pairs with hardware ext
AI2-THOR	Maintenance mode (2022)	pip	Light	fallback
CARLA	Active	Ubuntu 24.04 unsupported; ~20 GB	Heavy	✗ skip
Flightmare / Colosseum	Unmaintained / niche	—	—	✗ skip

- **Habitat-Sim** — chosen for the navigation extension.
  - Photorealistic real-building scans; the all-Python step API returns RGB, metric depth, and GT pose, so we inject our own mapper and plan with A* on our own Bird's Eye View (BEV) grid.
  - Built-in `GreedyGeodesicFollower` oracle = free baseline; Success Rate + SPL are citable metrics (Habitat, ICCV 2019).
  - ReplicaCAD scenes work day 1 with no approval; submit the free HM3D request immediately as backup.
- **Gazebo + ROS 2 Jazzy + Nav2** — the robotics-stack alternative.
  - Nav2's costmap accepts PointCloud2 from any topic: our perception node becomes a simulated TurtleBot 4's only obstacle sense (LiDAR disabled).
  - Chief appeal: the identical node drives the real-camera hardware demo. Chief risk: TF/QoS/costmap debugging concentrated on whoever knows ROS.
  - See darshmenon/rosnav under References — an active Nav2 stack that could jump-start this path.
- **AI2-THOR** — pip-install fallback if Habitat's conda/EGL install fights back; synthetic Unity homes; in maintenance mode since 2022.
- **CARLA** — great ground-truth-depth data generator, expensive navigation platform; Ubuntu 24.04 is unofficial and it is by far the heaviest option here (~20 GB install, GPU-hungry).

Challenges

Challenge	Venue / years	Task	Status for us (2026-07-21)
RoboWorld	NeurIPS 2026	RGB-D safe/social navigation	**Active**; deadlines ~Oct — post-course entry option

Challenge	Venue / years	Task	Status for us (2026-07-21)
CMU VLN	IROS 2026	Language-grounded navigation	Registration closes Jul 25; submission Aug 25 — post-course only
Habitat Challenge	CVPR EAI 2019–2023	PointNav / ObjectNav	Ended; **frozen benchmark reusable** — we adopt SR/SPL
BARN	ICRA 2022–2026	Cluttered ground-robot navigation	2026 concluded; scoring formula reusable
DodgeDrone	ICRA 2022	Vision-based drone obstacle avoidance	One edition; starter kit public
AI Driving Olympics	NeurIPS 2018–2021	Duckietown lane + obstacles	Dormant — the closest NeurIPS collision-avoidance challenge
Waymo / Argoverse / nuScenes / CARLA	CVPR WAD yearly	Driving perception suites	No 2026 cycles; leaderboards open year-round
Earth Rover	IROS 2024–2026	Real sidewalk-robot navigation	Active + free — event runs after the course
3LC Multi-Vehicle Detection	Kaggle community, 2026	2D vehicle detection (frozen YOLOv8n, label cleaning)	**Finished June 2026**; 2D-only — not useful for our eval
iGibson / RoboTHOR / MultiON	CVPR EAI 2020–2023	Interactive / object navigation	Dormant; benchmarks public

- **RoboWorld (NeurIPS 2026)** — the one live challenge aligned with our topic (Tracks 2/3/5: RGB-D perception + collision-free navigation among moving humans); our idea-6 variant is deliberately shaped as a post-course on-ramp.
- **CMU VLN** — the sensor suite (3D LiDAR + 360° camera) and VLN task sit a step away from ours; only the free registration (deadline Jul 25) matters, and only if we want the post-course option.
- **3LC (Kaggle)** — verified: a data-cleaning competition on UA-DETRAC traffic-camera footage with a frozen YOLOv8n; finished, monocular 2D only. Considered and set aside.

References — Repos, Tools & Papers

Repo / tool	What we'd use it for
ultralytics	YOLO11 detection, ONNX export, optional fine-tuning; also serves YOLO-World for idea 7
Depth-Anything-V2	Metric monocular depth (outdoor / indoor checkpoints); run PyTorch — the metric ONNX export has a known bug
kitti-object-eval-python + distance fork	KITTI 3D AP + distance-binned centroid error
nuscenes-devkit	nuScenes loading, transforms, official center-distance eval
VoteNet	SUN RGB-D 10-class eval protocol + code; pretrained learned baseline
habitat-sim / habitat-lab	Navigation-extension platform + oracle baseline
pyrealsense2	RealSense `.bag` recording + aligned-depth replay

Repo / tool	What we'd use it for
py-motmetrics	MOTA/IDF1 scoring for the tracking variant
processed_sunrgbd (HF)	MATLAB-free SUN RGB-D labels
darshmenon/rosnav	Active ROS 2 Nav2 navigation stack (Humble/Jazzy, last push Jul 2026) — jump-start for the Gazebo/hardware path
Real-Time-Object-Detection-for-Collision-Avoidance	One-off 2024 student project (YOLOv11 on KITTI 2D, inactive); only its KITTI→YOLO label conversion is a useful reference

- **Core papers to cite in the report** (full venue-verified list with links in papers.md; the professor requires true peer-reviewed venues — note LMDepth and ZoeDepth are arXiv-only):
  - Pipeline lineage: Frustum PointNets (CVPR 2018) — 2D boxes + depth frustums → 3D, exactly our shape; Pseudo-LiDAR (CVPR 2019) — depth map → point cloud is what makes camera 3D detection work; RANSAC (Comm. ACM 1981); occupancy grids (Elfes, IEEE Computer 1989); Stixel World (DAGM 2009); ground segmentation + clustering archetype (Zermas et al., ICRA 2017).
  - Depth: Depth Anything V2 (NeurIPS 2024) — our depth model; Metric3D (ICCV 2023) — why zero-shot *metric* depth is possible; MiDaS (TPAMI 2022); Depth Pro (ICLR 2025) — alternative backbone; RealSense sensor accuracy study (IEEE Access 2024) — published error-vs-distance curves to compare against our tape measurements.
  - 3D detection + datasets: MonoDLE (CVPR 2021) — localization error dominates, motivates our metric; image-based 3D detection survey (TPAMI 2024); ImVoteNet (CVPR 2020); KITTI (CVPR 2012); SUN RGB-D (CVPR 2015); nuScenes (CVPR 2020) — source of the center-distance protocol.
  - Navigation + variants: Habitat (ICCV 2019) — SR/SPL protocol; VLFM (ICRA 2024) — zero-shot semantic navigation, the template for idea 7; SORT (ICIP 2016) + DeepSORT (ICIP 2017) for idea 6; YOLO-World (CVPR 2024); BARN benchmark (IEEE SSRR 2020).
  - Camera-to-BEV lineage: Lift-Splat-Shoot (ECCV 2020), BEVFormer (ECCV 2022), SparseOcc (ECCV 2024) — the learned counterparts our geometric BEV map is positioned against; plus two close system peers: indoor obstacle discovery on reflective ground (IJCV 2024) and real-time monocular occupancy-grid mapping (J. Field Robotics 2026).
  - Components: YOLO (CVPR 2016) + YOLOv10 (NeurIPS 2024) — the citable detector pair (Ultralytics v5/v8/v11 have no peer-reviewed papers; cite the repo as software); A* (IEEE TSSC 1968); Kalman (Trans. ASME 1960); Hungarian method (NRLQ 1955).